

CSE370 – Database Systems
Assignment 04
Summer 2023

Suppose you have the table below for “Instructors” for a School Database with columns ID, Name, Subject and Salary respectively:

76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
45565	Katz	Comp. Sci.	75000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000
12121	Wu	Finance	90000
76543	Singh	Finance	80000
32343	El Said	History	60000
58583	Califieri	History	62000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
33465	Gold	Physics	87000

1. To speed up access to the data, you want to create an Index file using B+ tree with the Name column as the search key value. Construct a B+ tree of order $n = 3$. The search key values are inserted in the same order as shown in the above table. Show the expansion of the tree for each insertion and the final tree. [4]
2. Suppose the number of buckets in the hash index is 5 and each bucket can hold 3 index entries at max. Construct a hash index on attribute ID. Bucket overflow is handled using forward chaining. The hash function is as below: [3]

$$h = (\text{First digit of ID}) \% 5$$

3. Suppose the number of buckets in the hash index is 5 and each bucket can hold 2 index entries at max. Construct a hash index on attribute ID. Bucket overflow is handled using forward chaining. The hash function is as below: [3]

$$h = (\text{Sum of all digits in Salary}) \% 5$$